

Occurrence of Invertebrate Fauna in Schirmacher Oasis, Antarctica

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Abstract

The Schirmacher Oasis region in East Antarctica was surveyed during the eleventh Indian expedition to Antarctica in the austral summer of 1991-92. Samples of water, soil, moss, lichen and algae showed presence of invertebrate fauna like earthworm, nematods, mites, springtail, dipterous flies, larvae and a single specimen of moth.

Introduction

Terrestrial animals of Antarctica are confined to microbes and a few small invertebrates. A total of 387 invertebrate fauna have so far been recorded in the sub-Antarctic, 127 in maritime Antarctica and 107 on the continent itself; many of these occurring in more than one of these zones (May, 1988 and Laws, 1989). These are mostly microscopic, which live in the plant litter, moss, peat, soil, lichen, algae, birds' litter and in the micro-climate under the rocks. They range from protozoa, rotifers, tardigrades and nematodes to arthropods. Out of 137 species of Arthropods reported from Antarctica, the principal Arthropod groups represent Acarina (70 species of mites), Mallophaga (37 species of biting lice), Collembola (20 species of springtails), Anoplura (5 species of sucking lice), Diptera (2 species of midges), Siphonoptera (1 species of fleas) and Coleoptera (2 species of beetles probably alien or known from islands near the Antarctic Peninsula).

Ingole et al. (1987) have reported 14.57 % nematode 0.54 % acarina, 22.31 % protozoa, 0.60 % oligochaeta and 38.6 % resting stages of micro-invertebrates from the aquatic micro fauna of Priyadarshini Lake of Schirmacher Oasis. Ingole and Parulekar (1987) reported *Teratacephalus* sp. of nematode from water-moss communities at the same location. So far unidentified mite and springtail have also been reported from Schirmacher Oasis (Richter et al., 1989).

Materials and Method

Samples of water, soil, lichen, algae, moss, and birds' litter were collected from the different lakes existing all along the length of Schirmacher Oasis. Soil samples were also collected from the crops being grown inside the living rooms of Russian station Novolazarevskaya, glasshouse of German station Georg Forster and Indian station Maitri - all located in Schirmacher Oasis. Mosses, algae and lichens were collected from dried and moist bank of lakes and from running stream. The temperature of the site of collection was recorded. All samples were extracted through Berlese funnel and examined under the binocular microscope at Maitri station. Living samples of Dipteran flies from Novolazarevskaya and single moth from Maitri were collected and preserved. The collection of samples was done from last week of December 1991 to 15 February 1992. 100 gm of freshly collected soil samples were examined under binocular microscope. The moss, lichen and algae (50 gm each) were put in 50 ml of water and the samples were teased out with the help of forceps and needle in porcelain dishes separately. Water with sample from each dish was taken out with the help of dropper and kept in cavity block with black surface for examination under the binocular microscope. Specimens were collected and preserved in 70 % alcohol in plastic tubes. Detailed taxonomic studies are in progress.

Results and Discussion

The details of specimens collected, their host and site, are described below:

Earthworm: The living specimens of earthworm were collected from the soil sample taken from tomato and cucumber crop being grown in wooden frame inside the living hut at Novolazarevskaya. Specimens were first observed on 20 January 1992. These were clear white in colour and the size varied from 2 to 14 mm (Fig 1). Their number counted from 11 to 47 per 100 gm weight of soil sample. Earthworms are reported only from islands of sub- Antarctic region (Laws, 1989). Thus, this is the first reporting of earthworm from East Antarctica.

Nematode: The living transparent and white nematodes (Fig 2) were observed moving spirally in vertical position when the teased out sample of moss was examined under the binocular microscope. Nematode was collected only in moss sample from dried banks of the lakes. These were not found in the running water moss community. The variation in nematode population varied from 42 to 79, 46 to 88 and 27 to 43 in the months of December 1991, January 1992, and February 1992, respectively in 20 gm weight of moss. First collection of nematode was done on 28 December 1991. Ingole and Parulekar (1987) have

PLATE 1



Fig 1 : Earthworm (1x16x10X) collected from soil of Russian station in Antarctica.



Fig 2 : Nematode(1x35x10X) collectd from Antarctic moss

PLATE 2

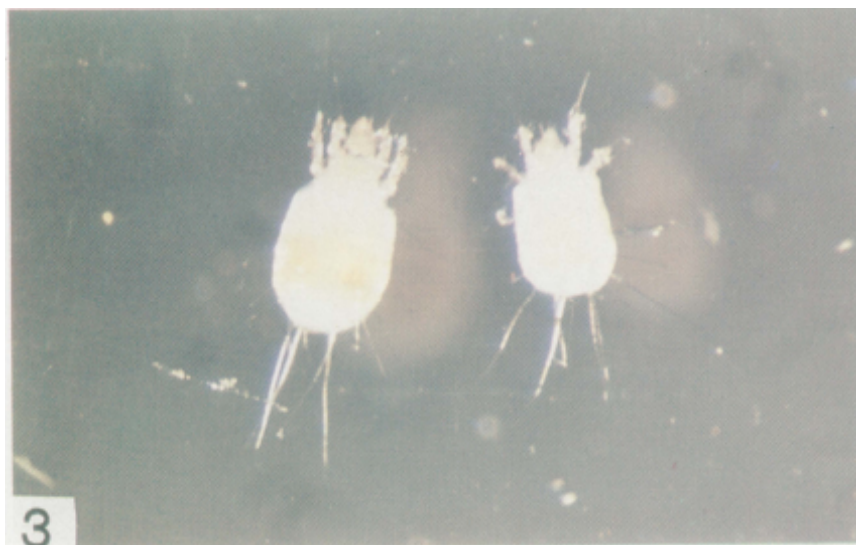


Fig 3 : Mite (1x30x10X) collected from Antarctic soil.



Fig 4 : Mite (1x30x10X) collected from Antarctic soil of Russian station (shown by arrow)

reported *Teratacephalus* sp. of nematode from the water-moss community of Schirmacher Oasis and Ingole et al.(1987) have noticed an unidentified nematode species in the aquatic microfauna of lake.

Mite: Two types of white mite (Figs 3 & 4) from the soil of cucumber, tomato and capsicum, one pink mite (Fig 5) from tomato soil of Novolazarevskaya and one white mite from the soil of maize crop of Maitri were collected. These four types of mite are quite different from each other. The mite (Fig 6) collected at Maitri is white, oval with dorsally convex body. Anterior portion is pointed and two pairs of anterior legs are projected forward. This type of mite is sluggish in movement and their population varied from 7 to 76 in 100 gm soil. The white mite from Russian station are of two types, one being like maize seed in structure with fine bristles projected outside the body (Fig 3). The population of this type of mite varied from 21 to 56 in 100 gm soil sample. The other type of mite is dorso-ventrally flattened, very strong with strong legs (Fig 4) and population counted was 3 to 11 in 100gm soil sample. These specimens were first collected on 5 January 1992.

Acarina represents the principal Arthropod group and the mites are large and relatively well known group in the region. About 70 species of mites are known from Antarctica which include feather mites on birds and a specialized group which live in the nasal passage of seals. Predator mite i.e. the fast moving yellow mite (*Gamasellus racovitzii*) and pink soil mite i.e. the most southerly (nearest to south pole) living animal (*Nanorchestes antarcticus*) also occur in Antarctica (Laws,1989). The *N. antarcticus* is barely visible being half to 1/3 of a mm long, is similar to the pink mite of present study. However, it needs confirmation after taxonomic studies. One unidentified mite has been reported from Schirmacher Oasis by Richter et al.(1989).

Springtail (Collembola): Two types of springtails were collected from the soil sample, the first being larger with average length of 1.25 mm (Fig 7) and the second type is small with average length of 0.65 mm (Fig 8). Both are white in colour. The larger type has slender body and actively jumps from one place to other whereas smaller type is dorso-ventrally flattened with broad oval abdomen and moves slow to hide in the soil. The body segments are more obvious in larger type. The population of larger and smaller types of springtails were counted as 6 to 28 and 12 to 56, respectively in 100 gm of soil sample. These specimens were first collected on 5 January 1992. So far 20 species of springtails have been reported from the Antarctic continent (Laws,1989). Two unidentified species were also collected by Richter et al.(1989) from the terrestrial habitats on mosses, lichens and from moistened algal mats in sheltered position, of Schirmacher Oasis. The springtails of present study were

collected only from soil with micro-plants whereas other samples (lichens, moss and algae) did not have these animals.

Dipterous flies: One larger type from Novolazarevskaya and one each of smaller type of flies from Novolazarevskaya and Maitri were collected on 5 and 9 January 1992. The larger flies measure, on an average, 2.5 mm in length and 2.0 mm in width. They have black, blunt end abdomen (Fig 9). Thorax and head were greyish yellow and the eyes were bulging. Wing measured 2.75 mm in length. Smaller flies (Figs 10 and 11) are poor fliers and hide under the soil clods. The abdomen of this type of fly was tapering, pointed and greyish black in colour whereas the head and thorax were dark black. The antennae and abdomen had 11 and 7 segments, respectively. Average length of this type of fly was measured as 3 mm in males and 2 mm in female. Both types were collected from cucumber, tomato and capsicum plants in Novolazarevskaya whereas small type were collected from cucumber, onion and mint plants at Maitri. The population of large and small flies collected from Novolazarevskaya was counted as 16.5 and 83.5 per cent, respectively. Two species of midge have already been reported from the Antarctic continent of which one is the largest and wingless i.e. *Belgica antarcticus* grows to a maximum of 12 mm length (Shapley, 1985). The flies of the present study are first reported such animal from Schirmacher Oasis.

Larvae: Two types of larvae were collected from Schirmacher Oasis, one being larger with black head (Fig 12) was collected from the soil of cucumber plants in Novolazarevskaya and the other being small (Fig 13) collected from the moss samples of dried bank of lakes. The bigger sized larvae are dirty white with cylindrical abdomen and without legs. These are inferred as larval stage of big flies. These, however, did not populate and ultimately died in laboratory rearing. The smaller larvae, collected from moss, are creamy white in colour with segmented body and three pairs of legs. These larvae settled down when the moss sample was teased out in water. Larvae are very sluggish. The population of larger and smaller larvae counted 2 to 9 and 5 to 11 in 100 gm weight of soil and 20 gm weight of moss samples, respectively.

Moth: Greyish brown coloured single living moth was collected from the wooden living cabin of Maitri on 2 January 1992. The moth was kept with sugar solution in jar which survived for four days. It measured 7 mm in length (head and thorax 2 mm and abdomen 5 mm) with 3.5 mm width. Fore-wing was measured 6.5 mm in length. Abdomen is blackish white whereas head and thorax are black (Fig 14). So far the lepidopterous insect has not been reported from the Antarctic continent.

PLATE 3

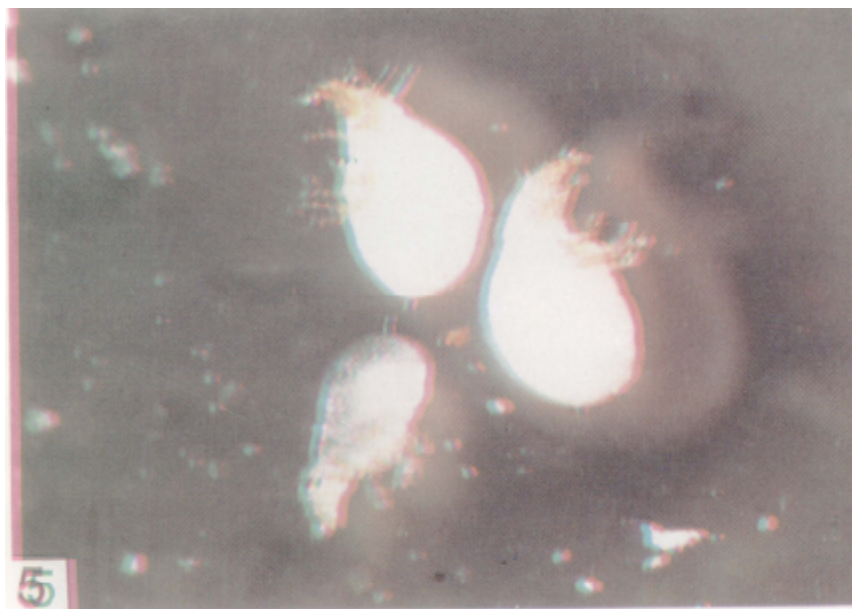


Fig 5 : Mite (1x30x10X) collected from soil in Maitri area.

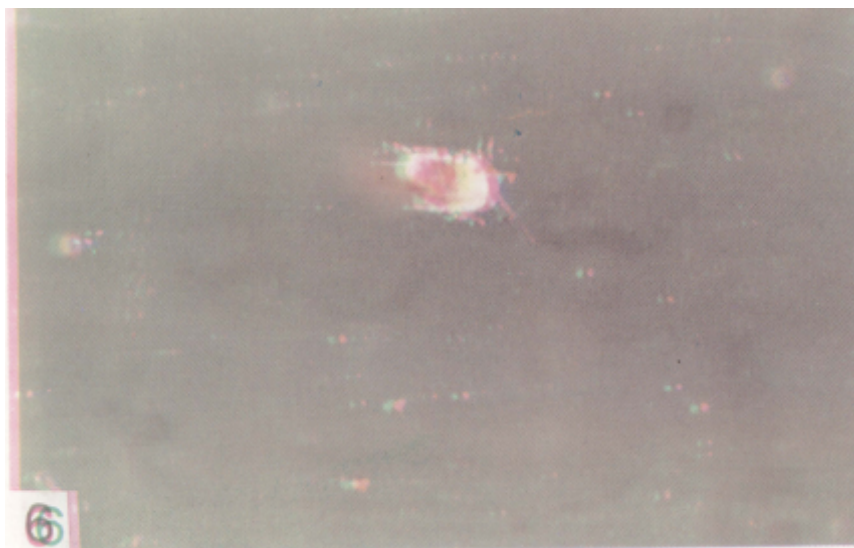


Fig 6 : Pink Mite (1x 35x 10X) collected from tomato soil of Russian station.

PLATE 4



Fig 7 : Large size of springtail (1 x 35x 10X) collected from soil of Russian station



Fig 8 : Small size of springtail (1 x 38x 10X) collected from soil of Russian station

PLATE 5

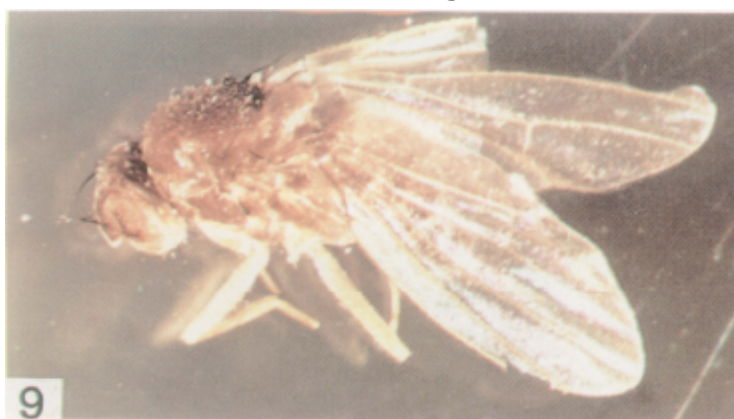


Fig 9 : Large fly (1 x 16 x 10X) collected from cucumber plant grown at Russian station.



Fig 10 : Small fly (1 x 16x 10x) collected from Russian and Indian stations.



Fig 11 : Small fly (1 x 16x 10X) both male(M) and female(F)

PLATE 6



Fig 12 : Big larva (1 x 16 x 10X) collected from soil of cucumber plant grown at Russian station.

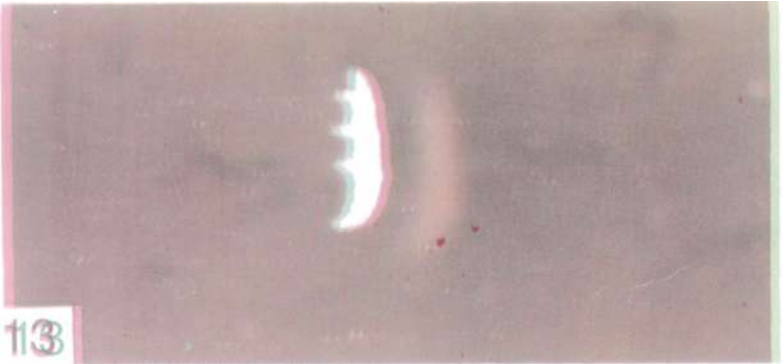


Fig 13 : Small larva (1 x 35 x 10X) collected from the moss of lakes in Schirmacher Oasis.



Fig 14 : Moth (1 x 10 x 10X) collected from wooden cabin of Indian station

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References

- Atlas Antarktiki,11(1969): Cidrometeorologicheskoe Izdatel'stvo, Leningrad (in Russian),pp 598.
- Bhattacharya,B.-B.(1987): Schirmacher Oasis : Environment history and the Indian station Maitri. Sci.Rep.fourth Indian expedition to Antarctica,Tech.Pub.No.4,Deptt.of Ocean Development,pp 171- 179.
- Ingle,B.S. and Parulekar,A.H.(1987): Microfauna of Schirmacher Oasis,Antarctica : 1. Water-moss communities. Sci.Rep.fourth Indian expedition to Antarctica,Tech.Pub.No.4,Deptt.of Ocean Development, pp 139-148.
- Ingle,B.S.,Verlencar,X.N. and Parulekar,A.H.(1987): Microfauna of Priyadarshini Lake,Schirmacher Oasis, Antarctica.Sci.Rep. fourth Indian expedition to Antarctica, Tech.Pub.No.4,Deptt. of Ocean Development,pp 149-154.
- Laws,R.(1989): Antarctica : The Last Frontier, British Library Cataloguing in Publication Data, Italy by OFSA S.P.A. for Boxtrene Ltd.,London.
- May,J.(1988): The Green Peace Book of Antarctica (A new view of the seventh continent). Dorling Kindersley Ltd.,London.
- Richter,W.,Haendel,P.and Jughans,P.(1989): The animals of the Schirmacher Oasis (East Antarctica). Ant.Res.,Proc.Symp.held at Potsdam GDR, 11-14 Sept.,1989,V II,pp 495-503.
- Shapley,D.(1985): The Seventh Continent Antarctica in a Resource Age. Published by the Resources for the Future, P.Street,Washington.